MARKET ENTRY RESEARCH DOSSIER

Chinese electric two-wheelers into Pakistan

A go/no-go investigation built from audited company filings, Pakistan's primary tariff law read in its own words, and a working cost model. Compiled to survive scrutiny, including where the finding is unwelcome.

Compiled 27 August 2026

Prepared for Sheheryar Asif

Desk research, plus five supplier RFQs sent 27 Aug 2026

No bank, delivery platform or regulator contacted

Evidence grades used throughout:

AUDITED company filing ASSOCIATION industry body PRESS unverified DECLARED self-reported

Contents

TWO STANDING RULES THIS DOSSIER FOLLOWS

No figure appears without a source and an evidence grade. Where a company filing and the trade press disagreed, the filing won — and in one case that reversed a conclusion already drawn.

No currency is converted without a dated rate. Figures stay in their source currency; comparisons are structural.

SECTION 1

Verdict

4–6	7×	88%	30 Jun 27
MONTHS PAYBACK · FLEET · NO SUBSIDY	CHEAPER PER KM THAN A CD 70	MARKET HELD BY ATLAS HONDA	TARIFF CONCESSIONS SUNSET

The economics of electric two-wheelers in Pakistan are **outstandingly good**. They survive the worst domestic electricity tariff, a punishing battery assumption, and the complete removal of every government incentive — simultaneously. **Demand is not this project's risk.**

WHAT THE EVIDENCE SUPPORTS

A fleet-first entry serving delivery riders, sold on arithmetic. Fuel is **30–71% of a Pakistani delivery rider's gross earnings**. An electric bike repays its premium in **four to six months** with no subsidy. There are **50,000+ known fleet riders**, and PKR 50m buys enough machines to need **under 1% of that market**. It is the only thesis found that survives every constraint in this dossier.

WHAT THE EVIDENCE DOES NOT SUPPORT

A new consumer brand. Atlas Honda holds ~88% of the market on 1.69m units a year, runs 500+ dealers and 63 service points, and has launched an electric scooter while stating it needs *no additional investment* to do so. India ran this experiment: the best-funded pure-play fell from **over 50% share to 6.8%** — beaten on service, not price.

THREE GATES CANNOT BE CLOSED FROM A DESK

Whether a manufacturer will supply at this scale (K3), whether a letter of credit can be opened today (K4), and whether an enforceable battery warranty backstop exists (K5). **Each is a conversation.** A fourth has been added by this research: whether a delivery platform will deduct rental from rider earnings at source — the mechanism the fleet model depends on.

SECTION 2

The demand case

Running cost — the most robust number in the dossier

Honda CD 70 at PKR 343.10/litre against an electric equivalent at 2.5 kWh/100 km, across Pakistan's domestic tariff slabs.

VEHICLE / TARIFF	PKR PER KM	NOTE
Honda CD 70 — petrol	**5.72 – 6.24**	55–60 km/l real-world
Electric — unprotected top slab 68.00/kWh	1.70	Worst domestic case
Electric — above-700 slab 47.20	1.18	
Electric — commercial 39.70	0.99	NEV Policy rate
Electric — national average 33.38	**0.83**	NEPRA SRO 279(I)/2026

Even at the most expensive domestic electricity in Pakistan, the electric bike costs about a third of the petrol bike to run. At the national average it is roughly one-seventh. This depends only on the price of petrol and the price of a unit of electricity — not on a forecast, a subsidy or a policy.

PAKISTAN NOW REPRICES FUEL DAILY — VERIFIED 27 AUGUST 2026

The petrol figure was checked against a conflicting press number of 414–415 and **PKR 343.10 is confirmed correct**: the whole month ran 329.82 → 343.10, a 4.0% band, across four independently reported notification dates. **Nothing approaches 414.**

The larger discovery is structural. **Since August 2026 petrol is reviewed daily under a new OGRA mechanism** — the 26 August notification was valid only "until August 27, 2026." That replaces the fortnightly regime this research was built on. **No petrol figure here can be quoted as "the price" again, only as the price on a stated date.**

It cuts both ways commercially. **It strengthens the fleet pitch** — a rider facing a price that moves every morning values a fixed electricity cost more, and volatility is what an electric bike actually removes. **And it adds a risk**: a sustained crude fall now reaches the pump in days rather than weeks, eroding the saving faster than before.

Payback and its sensitivities

CASE	PREMIUM PKR	BREAKEVEN	MONTHS
Consumer, 15,000 km/yr, no subsidy	88,600	16,400–18,100 km	13.1–14.5
Consumer, with PKR 65,000 subsidy	23,600	4,832 km	**3.9**
Fleet, 24,960 km/yr, lithium	88,600	–	**5.9**
Fleet, 40,000 km/yr, lithium	88,600	–	**3.7**

Battery sensitivity: even at PKR 60,000 replaced every 15 months — a deliberately punishing assumption — the electric bike still saves PKR 1.69/km. Battery cost changes the payback period; it never reverses the sign.

THE SLAB-CROSSING TRAP — SPECIFIC TO PAKISTAN

Charging draws roughly **31 kWh a month**. Pakistan's protected domestic tariff cuts off at **200 units**, and crossing it removes protected status **from the entire bill, not just the marginal units.** A household at 175–200 units — plausibly exactly where a CD 70 buyer sits — is pushed over.

The consequence is commercial: **a naive savings pitch is contradicted by the customer's next electricity bill.** It argues for charging away from home, and makes **solar charging a genuine differentiator** rather than a marketing line, because it bypasses the slab entirely.

SECTION 3

The fleet model

The rider

PARAMETER	VALUE
Duty cycle	**80 km/day, 26 days/month** — ~24,960 km/year
Efficiency	35–50 km/l on a 125cc; 40 km/l used
Reported fuel spend	~PKR 800/day PRESS
Earnings	PKR 25,000–60,000/month
Known fleet size	foodpanda 20,000+ · Bykea 30,000+

Cross-check: 80 km ÷ 40 km/l × PKR 343.10 = **PKR 686/day** computed against ~PKR 800/day reported — close enough to validate the model.

THE FINDING

Fuel is PKR 17,841 a month — between 30% and 71% of a delivery rider's gross earnings. Electricity for the same distance is **PKR 1,736.** This is not a lifestyle argument; it is most of a working person's income.

Cycle life replaces calendar life

At 80–128 km/day against a 75–85 km range, batteries cycle **312–500 times a year**. Lead-acid, rated 300–400 cycles, therefore lasts **8–13 months** in fleet service.

Both chemistries pay back in roughly the same time, because lithium's higher cost is offset by its life. **So chemistry is not an economic choice here — it is an operational one**, and getting it wrong is precisely how India's largest player lost its market.

Rider affordability — solved by structure

A rider earning PKR 35,000/month cannot produce PKR 248,500 upfront. **It was never an affordability problem; it was a financing-structure problem.** The rider does not need savings — they need a payment smaller than what they already spend on petrol.

RATE/DAY	RIDER KEEPS/MONTH	OPERATOR COLLECTS/MONTH	PAYBACK @ PKR 150K BIKE
PKR 300	8,305	7,800	19.2 mo
PKR 350	**7,005**	**9,100**	**16.5 mo**
PKR 400	5,705	10,400	14.4 mo
PKR 450	4,405	11,700	12.8 mo

Headroom created by replacing petrol with electricity: **PKR 16,105/month = PKR 619 per working day.** Any charge below that leaves the rider better off from day one, with no deposit, savings or credit history.

THE COLLECTION MECHANISM IS WHAT ACTUALLY SOLVES IT

Renting assets to informal workers is a credit-risk business, and that is usually where this model dies. **The fix is to collect at the platform, not from the rider** — foodpanda or Bykea deducting the daily rate from earnings before payout. That converts thousands of informal-worker credit exposures into **a single business receivable from a funded company.**

Proven template: Zypp Electric in India runs exactly this — ₹100–300/day to delivery riders converting to ownership at 52 weeks, 20,000+ vehicles across Swiggy, Zomato, Blinkit and Uber riders, on US$76.5m raised. Same duty cycle, same constraint, same customer.

WHAT IT TRADES AWAY

Rental **locks capital for 13–20 months per machine** where a sale recovers it immediately. In exchange: recurring revenue, the battery margin, and control of uptime. **This becomes an asset-finance business with a supply arm, and should be funded as one.**

SECTION 4

The incumbent

1.69m	~88%	500+	63
UNITS SOLD FY26	MARKET SHARE	SALES DEALERS	SERVICE & PARTS POINTS

Pakistan's motorcycle market is effectively a monopoly. Atlas Honda's network is 38 3S, 20 2S and 5 1S outlets plus 500+ sales dealers, with stated capacity above 1.5m units a year. Other players — United, Road Prince, Yamaha, Suzuki, Qingqi, Ravi — share the remaining 12–15%.

Road Prince matters disproportionately: one of the few substantial non-Honda networks, and it already carries Yadea, the world's largest electric two-wheeler maker.

Honda ICON e: — what the incumbent's electric scooter actually is

SPECIFICATION	VALUE
Price	**PKR 419,900**
Motor	1500 W · 85 Nm · 55 km/h

SPECIFICATION	VALUE
Battery	**Li-ion 1.58 kWh — FIXED, non-swappable**
Range	**65 km**
Charging	**6.0 hours · proprietary charger · no fast charge**
Weight / wheels	86 kg dry · 12″

THIS IS THE MOST CONSEQUENTIAL FINDING FOR THE FLEET THESIS

At PKR 419,900 the ICON e: is **2.6× a CD 70** and **1.7× a Yadea T5**, while offering **less range than the T5** (65 km vs 75–85 km).

And it is structurally useless for fleet work. A rider covering 80–128 km a day cannot use a 65 km machine with a fixed battery, a proprietary charger and a six-hour recharge. No swap, no fast charge, no second pack. **The incumbent has taken a premium consumer position, not a fleet one — the gap is confirmed open by the only company that could have closed it.**

SECTION 5

India — the market that already ran this cycle

Same subcontinent, same commuter baseline, same price sensitivity, and a subsidy structurally similar to Pakistan's NEV cost-sharing. India ran the full cycle — boom, withdrawal, fraud investigation, shakeout — between 2022 and 2026. **Pakistan is at the start of that curve.**

PERIOD	WHAT HAPPENED
2022	Generous FAME-II support draws in entrants. **Hero Electric and Okinawa hold 31% combined share.**
2023–24	SFIO pursues **Hero Electric, Okinawa and Benling** over ~INR 2.97bn in localisation-linked claims. **~INR 469 crore ordered repaid; 80%+ of permanent staff laid off.** Combined share falls 31% → 10%.
2024	FAME-II lapses. **Sales fall 53% in a single month.** Beneficiary vehicles cut from ~989,000 to under 564,000.
2024–26	Ola Electric falls from **>50% share to 6.8%**. 10,644 consumer-helpline complaints, warranty claims refused, scooters queued for parts — **Goa suspends its vehicle registrations.** FY loss ₹1,833 crore.
2026	Registrations **up 53–75%** with support slashed to ₹5,000/vehicle. **TVS 25%, Bajaj 23%, Ather 16%, Hero 11%** — those four take **95.6% of incremental growth.**

THE TRANSFERABLE LESSONS

1. Demand survives subsidy withdrawal. Entrants often do not. The market was captured by whoever already had dealers, service and brand trust. In Pakistan that means Atlas Honda — and it has begun.

2. Service is the decisive variable, not price. Ola had capital, product, brand and half the market. It lost on spare parts and service capacity — and a state regulator removed its right to sell.

3. Localisation-linked subsidy claims carry clawback risk. Pakistan's 1% duty concession is gated on EDB certification and annual input reconciliation. Same structural shape. In India, getting that paperwork wrong was company-ending.

SECTION 6

Supply side

Audited financials — read from the filings AUDITED

COMPANY	REVENUE	NET MARGIN	VEHICLE UNITS	INTERNATIONAL
Yadea 1585.HK · FY2025	RMB 37.01bn	**7.87%**	16,269,200	Vietnam plant live
Aima 603529.SH · FY2025	RMB 25.10bn	**8.11%**	11,639,817	**0.86%** ↓7.89%

COMPANY	REVENUE	NET MARGIN	VEHICLE UNITS	INTERNATIONAL
Luyuan 2451.HK · FY2024	RMB 5.07bn	2.29%	–	HKD 6.1m unspent
Xinri 603787.SH · FY2025	RMB 4.11bn	1.23%	2,284,597	2.48% @ 9.95% GM
NIU NASDAQ · FY2025	–	**loss**	1,192,039	6.7% of units

THESE ARE DOMESTIC CHINESE BUSINESSES WITH AN EXPORT GESTURE ATTACHED

Aima — described everywhere as aggressively regionalising — filed **overseas revenue of 0.86% of turnover, and it fell 7.89%. Xinri** exports at 9.95% gross margin against 14–21% domestically: exporting is its *least* profitable channel. **NIU** sold 80,018 units internationally across 40+ countries — about 2,000 per market. **Luyuan's** international budget was HKD 6.1m, "delayed and unutilized".

A partner for whom export is under 1% of revenue and falling will not fund your warranty claims, marketing or credit terms.

The cost basis, and the MOQ chasm

Derived from Yadea's audited revenue table across 16.3 million units — **ex-works ASP: RMB 1,415 per electric bicycle, RMB 1,878 per electric scooter.** Chinese domestic retail for a mainstream 60–80 km machine is **CNY 2,999–3,899**, so domestic distribution margin is thin.

Order minimums split the market in two. Huaihai — a top-five maker with its own OEM arm, and an existing Pakistan base — quotes **10,000 units a year**, roughly twenty times the capital ceiling. The tier that deals at **50–200 units** is trading houses and small ODMs. **Nothing sits in between**, and the reachable tier is where a warranty backstop is weakest.

The battery layer

CHEMISTRY	PACK PRICE PKR	RATED CYCLES	CONSUMER LIFE	FLEET LIFE
Lead-acid / graphene	10,000–20,000	300–400	**12–15 months**	**8–13 months**
Lithium-ion	40,000–120,000	2,000+	3–5 years	3–5 years
NiMH	25,000–50,000	–	–	–

Batteries are the business, not the machine. Yadea's audited accounts show batteries and chargers at **28.4% of revenue — RMB 10.5bn** — and it sold **18.3 million batteries against 16.3 million vehicles.** The market leader *monetises* battery replacement rather than absorbing it as warranty cost.

The like-for-like lithium premium is now measurable in Pakistan: **MS Jaguar Bolt at PKR 174,000 standard versus PKR 245,000 in LFP — +PKR 71,000 for the same machine.** REVOO shows +42,000 on the A11 and +62,000 on the A12.

THE DEVELOPMENT TO WATCH

Yadea's sodium-ion Q1 and DE3 at CNY 3,299–3,499 — its entry models now use **no lithium at all**, with 92% capacity at −20°C and 80% charge in 15 minutes. If sodium reaches export markets it resets the cost floor and removes lithium price and supply risk from the bill of materials. **It is the market leader deploying it, not a startup.**

SECTION 7

Regulation

ITEM	POSITION
Tariff line	**PCT 8711.6040** — motorcycles with electric propulsion
EV-specific CKD components	**1% customs duty** — seven named components, read from the primary text
Non-localised CKD parts	**15%** — *corrected;* the widely-quoted "10–15%" is FBR's summary and is wrong
Localised CKD parts	**15% + Additional Customs Duty** under SRO 693(I)/2006 — **dearer than non-localised, by design**

ITEM	POSITION
Legal basis	Table-II to Part-V(A), **Fifth Schedule to the Customs Act 1969** — under **EV Policy 2020**, not the NEV Policy 2025-30
Expiry	**30 June 2027** — FBR's own word is "sunset"
Successor policy	AIDEP 2026-31 — **unapproved draft**, awaiting the PM and the IMF
NEV subsidy	PKR 65,000/two-wheeler, tapering to 80/60/40% in years 3/4/5
Installed local capacity	**~2 million units/yr, "heavily underutilized"** — the government's own words

THE GATE BEHIND THE RATE IS WRITTEN FOR A DIFFERENT TECHNOLOGY

The 1% is not automatic — it requires **certification and quota determination by the Engineering Development Board**, measured against SRO 656(I)/2006. **That instrument contains no electric-vehicle provisions at all**; its only use of the word "electric" describes a crane hoist.

The motorcycle schedule an e-bike assembler is judged against mandates an **engine assembly line** — crank cases, crankshaft and connecting rod, spark plug and tappet adjustment, oil filling — plus an **emission tester** at final inspection. **An electric motorcycle has none of those.** Certification therefore rests on how EDB applies an obsolete schedule; eligibility, cost and timeline cannot be self-assessed before applying.

Also required: Sales Tax Act 1990 registration; on-site facility verification; component lists approved by EDB or IOCO and uploaded to WeBOC; and **annual input reconciliation by 15 August, failing which the manufacturing certificate is not revalidated.**

One on-ramp, status unconfirmed: AIDEP allowed 10 built-up units per variant at half duty, to a maximum 200 units per company, with manufacturing compulsory within two years. AIDEP expired 30 June 2026 and it is **unconfirmed whether the two-wheeler pilot continued.**

The seven components that get 1% — from the schedule itself

Part-V(A), Table-II, serial 3 — **Electric motorcycle (PCT 8711.6040)**, read from the FBR Fifth Schedule rather than a budget summary.

	COMPONENT	PCT CODE
(a)	Electric Motor	8501.3200
(b)	Battery Charger	8504.4020
(c)	Switches	8536.5029
(d)	Junction Box	8536.4910
(e)	Controller	8542.3100
(f)	Converter	**8454.1000** ⚠
(g)	**Batteries other than lead acid**	**85.07**

A DRAFTING DEFECT IN THE SCHEDULE, AND IT IS THE KIND THAT ENDS COMPANIES

8454 is "converters, ladles, ingot moulds and casting machines of a kind used in metallurgy." It is a steel-making vessel. The auto-rickshaw entry two serials above gives Converter as **8502.4000** — electric rotary converters, the sensible code.

So the schedule assigns a **foundry code to an electric motorcycle's converter.** An importer declaring a DC-DC converter under 8454.1000 is declaring a steel converter. Whether Customs honours its own schedule or reclassifies is a question for a clearing agent before goods ship — **and paperwork exactly this size is what ended Hero Electric and Okinawa in India.**

The battery has three tariff routes, and the cheapest avoids EDB entirely

ROUTE	DUTY	WHAT IT REQUIRES
A · Pack as an EV CKD component "Batteries other than lead acid", 85.07	**1%**	**EDB certification as an electric motorcycle manufacturer** + quota

ROUTE	DUTY	WHAT IT REQUIRES
B · Lithium-ion cells 8507.6000, Part-III serial 143	**0%**	Sales Tax registration as a **lithium-ion battery manufacturer** + IOCO quota. **No EDB test in Part-III**
C · Full pack bill of materials 13 lines, cells to terminal covers, Part-I	**0%**	Sales Tax registration as a **lithium battery assembler** + IOCO quota, **and** Part-I's locally-manufactured-items test (EDB may certify)
E · Charging station for electric vehicle 8504.4030, Part-I	**0%**	**Condition column reads "Nil".** No EDB, no IOCO, no registration test, no expiry
No qualification	**10% + 2%**	CD 10% + RD 2%, then ST 18% and withholding 11%

TWO FINDINGS THE PRIMARY TEXT PRODUCED THAT THE SUMMARY NEVER SHOWED

Lead-acid is expressly excluded from the concession. The words are "batteries other than lead acid," in all three two- and three-wheeler serials. **Pakistan's tariff will not subsidise the cheap chemistry** — which happens to be the chemistry that fails in 8–13 months at fleet duty. Policy and engineering point the same way here.

And the state has built a lower-barrier on-ramp into the battery than into the vehicle. Neither route B nor C needs EDB certification as a *vehicle manufacturer* under SRO 656(I)/2006 — the heavy gate, and the one route A cannot avoid. Given the battery is **53% of FOB cost**, that is a materially different business to enter, at a materially lower regulatory bar.

Stated precisely, because the two routes are not identically clean: route B sits in Part-III, whose conditions are online information-furnishing only — **no EDB test whatsoever.** Route C sits in Part-I, which additionally requires the goods not be on the locally-manufactured list notified by Customs General Order, *or* be certified as such by EDB. **That is a far lighter test than vehicle certification, but it is not nothing** — if a casing or harness is on that list, the line loses its 0%.

And the cleanest concession in the whole schedule is the one nobody looked for: a charging station for an electric vehicle enters at **0% with "Nil" in the condition column.** No certification, no quota, no expiry. Since uptime is the product for a fleet, and a 65 km machine on a six-hour charge is useless at an 80–128 km duty cycle, **the hardware that answers that problem lands duty-free.**

What it costs if you do not qualify

LINE	8507.6000 LITHIUM	8711.6040 E-MOTORCYCLE CBU
Customs duty	10%	**30%**
Additional customs duty	–	**0%** exempted, SRO 1063(I)/2026
Regulatory duty	2% (to 30 Jun 27)	–
Sales tax	18%	18%
Income tax withholding	11%	12%
Provincial cess, Karachi	1.80–1.85%	1.80–1.85%

And the chemistry is taxed differently — by a factor of two

LINE	DESCRIPTION	CD	ACD	RD	DUTY PRE-TAX
8507.6000	**Lithium-ion**	10%	–	2%	**12%**
8507.2000	Other lead-acid (the traction line)	20%	2%	2%	**24%**
8507.1010	Lead-acid starter, expressly *"meant for vehicles of heading 87.11"*	25%	11%	–	**36%**

POLICY AND ENGINEERING POINT THE SAME WAY, FOR ONCE

Pakistan penalises lead-acid twice, and the penalties compound: it is excluded from the 1% concession by name, and it carries **double the statutory duty** of lithium — triple on the starter line.

So the cheap Chinese graphene/lead-acid tier — **the one that dies in 8–13 months at fleet duty** — is also **the one the border taxes hardest.** The chemistry that survives Pakistani fleet service is the chemistry the tariff favours.

Source: Trade Information Portal of Pakistan, Pakistan Single Window. **No total is stated** because sales tax and withholding apply to duty-inclusive value and the stacking order is a clearing-agent question, not a desk one. A **PSQCA Release Certificate is a listed requirement on the battery line**, with non-conforming shipments liable to destruction. The **Pak-China FTA rate of 4.58% on 8507.6000 expired 31 December 2023** — assume no FTA preference.

THE PRIZE IS NOT THE 1%

The prize is not paying 30% + 4% + 18% + 12% on a built-up unit. That is the real spread the policy creates, and the gate to it is EDB certification measured against **SRO 656(I)/2006 — an instrument the primary text confirms is the operative condition, and which contains no electric-vehicle provisions whatsoever.**

So K7 and the EDB question are not two risks. They are one.

THE EXPIRY DATE, NOW SETTLED IN BLACK-LETTER LAW

The FY2025-26 schedule printed **"five years from 1st July, 2020"** — which reads as expired. **Finance Act 2026 s.3(14) substituted the entire Fifth Schedule**, and the substituted text at *Gazette of Pakistan, Extraordinary, 26 June 2026, page 769* replaces that wording with:

> *"The concession shall be admissible to manufacturers of electric motorcycle* ***on and from the 1st day of July, 2025 till 30th day of June, 2027*** *subject to certification and quota determination by the Engineering Development Board (EDB)."*

The four-wheeler entries on page 766 moved from 2026 to 2027 identically. **The date is no longer an inference from a budget summary.** But read what it means: **ten months remain**, and EDB certification, supplier selection, an LC, a 15–45 day lead time and a 25–35 day sea leg all have to fit inside it. **The decision window is materially shorter than the concession window.**

AND THE 4% ADDITIONAL DUTY IS NOT PAYABLE

SRO 1063(I)/2026 of 30 June 2026, superseding SRO 1151(I)/2025, levies additional customs duty and then exempts, at paragraph 3(xi), **"Imports under PCT codes 8703.8030, 8711.6040 and 8711.6060"** — electric auto rickshaws, electric motorcycles and 3-wheeler loaders. Effective 1 July 2026, **with no expiry attached.**

Paragraph 3(iii) also exempts **imports under the Fifth Schedule generally**, subject to named exclusions. **So CKD components at 1% carry no ACD either.**

SECTION 7C

The vehicle in law, and what it costs to put on the road

India's entry tier exists because India created a **25 kmph class beneath registration**. This checks whether Pakistan has one. Source: **Provincial Motor Vehicles Ordinance, 1965**, read in full.

NO LOW-SPEED CLASS EXISTS, AND THE LAW LEAVES NO ROOM TO READ ONE IN

"Motor vehicle" means any mechanically propelled vehicle adapted for road use *"whether the powers of propulsion is transmitted thereto from an external or internal source"*. **"Motor cycle"** adds only two tests: fewer than four wheels, and unladen weight not exceeding **900 pounds (408 kg)**.

There is no wattage, kW, cc or top-speed threshold anywhere in either definition — the whole Ordinance was searched. And section 23(1) is unconditional: *"No person shall drive any motor vehicle ... unless the vehicle is registered."*The exemptions that exist cover dealers, road-rollers and driving to the registration office. **Nothing keyed to power or speed.**

So **every electric two-wheeler in Pakistan is a motor cycle in law** — the heaviest in the catalogues, the 86 kg ICON e:, is under a quarter of the weight ceiling. **The sub-PKR-97,000 tier cannot be unlocked by regulatory arbitrage**; it would take amending primary legislation in four provinces separately.

⚠ **This corrects a claim repeated across Pakistani EV vendor sites** — that machines below 250 W may not need registration. **It has no basis in the Ordinance** and appears to be Indian rules restated for a Pakistani audience. Selling on it would hand the buyer a legal exposure.

How a zero-cc vehicle is assessed — and where the ambiguity sits

Every provincial fee schedule is banded by engine capacity. **Balochistan publishes the bridge:** fees are charged in the ordinary band *"after conversion of power of electric motor from Kilo Watts (KWs) equivalent to engine capacity (cc)."*

THE MECHANISM IS PUBLISHED; THE NUMBER IS NOT

No conversion ratio is given anywhere in the schedule. That is not a rounding detail — a 1,500 W motor could land either side of the 149 cc line, which in Sindh is the difference between **1% and 2% of vehicle value.**

PROVINCE	ELECTRIC VEHICLE TREATMENT	EVIDENCE
Balochistan	Motor vehicle tax **waived in full to 30 Jun 2030**; kW→cc conversion published; motorcycle token **Rs 1,500 lifetime**	SCHEDULE
Punjab	**95% exemption** on registration fee and annual motor vehicle tax	PRESS
KPK	No token tax on electric *four-wheelers* to 30 Jun 2028	PRESS
Sindh	**No published EV concession found.** Bands: ≤149cc **1% of value**, 150cc+ **2%**. Only EV mention is a **Rs 5,000 late-registration penalty on "EV Scooters"**	SCHEDULE

KARACHI IS IN THE ONE MAJOR PROVINCE THAT OFFERS ELECTRIC VEHICLES NOTHING

Three provinces publish concessions. **Sindh publishes a penalty.** Keep it in proportion — on a PKR 150,000 machine the fee is PKR 1,500–3,000, so across 200 machines the gap against Punjab is **PKR 300,000–600,000. Real, not thesis-changing.**

The signal matters more than the sum. Provincial support cannot be assumed in the market this business would actually operate in, and any pitch leaning on "government support for EVs" is weaker in Sindh than the national NEV Policy implies.

AND ONE OBJECTION THIS REMOVES

A delivery rider's machine was always going to be registered, licensed and plated. **The fleet model never depended on an exemption**, so none of this weakens it — it closes off a different strategy. Registration also makes every machine **traceable, insurable and pledgeable**, which is what makes a rental fleet financeable at all.

SECTION 8

Landed cost — CKD kit pricing

Published listing prices from Alibaba, displayed in PKR by the platform's own undated conversion. DECLARED **These are not quotes.** No supplier was contacted.

PRODUCT	PKR	MOQ
KeywayEV K037 — 12″, 72V, 1001–2000W, SKD for local assembly	**51,706**	50
CKD SKD 10″ — 600/800W, 50 km/h	**55,748**	50
SAIGE EV Group — EEC, 1500W, 48/60V, 45 km/h	57,617–59,174	150
CKD EEC 10″ — 600/800W	61,977	50
K025 — off-road, 72V, 1001–2000W	87,792	50
Benlg ROBIN S — CKD/SKD, 48V, 450W hub	84,400–93,121	30
EEC 72V 2000W streetbike (LVJIAN)	102,775–105,890	50
Factory direct 72V 3000W, 90 km/h	155,408	50
3000W kit INCLUDING lead-acid batteries	**348,189–354,418**	20

WHAT THIS ESTABLISHES

A first order of 50 kits at ~PKR 55,000 is PKR 2.75 million — 5.5% of a PKR 50m ceiling. Against Pakistani retail, where the cheapest bike is PKR 97,000 and the volume band is 150,000–250,000, **the import stack has real margin in it rather than assumed margin.** This is the first hard evidence of that.

The battery — priced separately, and it costs more than the kit

Converted at **USD 277.90**, interbank selling, dated **27 August 2026, 15:56 PST** (Forex.pk). Same nominal spec across three named suppliers.

SUPPLIER	SPEC	TIER	USD	PKR
Dongguan Rishengzhi	72V 30Ah LiFePO4	MOQ 500	151.20	**42,018**
Koyosonic Power	72V 35Ah LiFePO4 + BMS	50+	213.32	**59,281**
Koyosonic Power	same	30–49	225.54	62,677
Hunan CTS Technology	72V 30Ah LiFePO4/NMC	100+	363.00	100,878
Hunan CTS Technology	same	50–99	410.00	**113,939**
Hunan CTS Technology	same	1–49	463.00	128,668
Pakistani retail benchmark	72V 30Ah LiFePO4, 70 km	retail	–	**115,000**

Pakistani benchmark: Maaz Electronics, PKR 115,000 discounted from PKR 150,000 — and **out of stock**, itself a data point about local supply depth.

THE FINDING: THE SUPPLIER SPREAD IS BIGGER THAN THE IMPORT MARGIN

At the **50-unit tier** — the tier this capital base actually reaches — the same nominal pack is **PKR 59,281 from Koyosonic** and **PKR 113,939 from Hunan CTS.** One is 48% below Pakistani retail; the other **matches Pakistani retail before a single rupee of freight, duty or tax.**

The arbitrage is not a property of the trade. It is a property of the supplier. Choosing wrong erases the entire margin — which is the clearest possible argument that listing prices cannot substitute for quotes, and it applies to the kits too.

WHAT IT DOES TO THE STACK, AND THE VOLUME CLIFF

Cheapest kit **PKR 51,706** + best 50-unit battery **PKR 59,281** = **PKR 110,987 FOB before anything else.** The battery is **53% of FOB cost** — the suspicion flagged earlier, now confirmed with figures. Against retail of PKR 97,000 cheapest and 150,000–250,000 volume band, **the margin is real but far tighter than the kit price alone implied.**

And the cliff is steep: **PKR 42,018 at MOQ 500 against PKR 128,668 at 1–49 — 3.1× cheaper at volume.** But MOQ 500 is **PKR 21m of batteries alone, 42% of the ceiling**, before a single vehicle. **The MOQ chasm again, cutting the same way.**

ONE ASSUMPTION UPGRADED FROM GENERIC TO MANUFACTURER-SPECIFIED

Hunan CTS publishes **1,500 cycles at 90% DOD**. At the fleet duty cycle of 312–500 cycles a year, that is **3.0–4.8 years of fleet service** — confirming the lithium case from a specification rather than a generic claim.

Read conservatively it is **three years, not five** — and Koyosonic warrants only **1 year below 50Ah**, exactly the pack size a two-wheeler uses. **A one-year warranty on a three-year asset is precisely the K5 exposure.**

SECTION 8B

The multiple from factory gate to shelf

The first dated FX rate in this research unblocks the comparison every section was waiting on. **Thu 27 August 2026, 15:56 PST — USD/PKR 277.40 / 277.90, CNY/PKR 41.26 / 41.34** (Forex.pk, indicative interbank). Conversions use the selling side.

POSITION	PKR	× EX-WORKS
Yadea ex-works ASP, e-scooter AUDITED	77,636	1.0×

POSITION	PKR	× EX-WORKS
Yadea ex-works ASP, e-bicycle AUDITED	58,496	0.75×
Tailg promotional, Chinese retail	123,978	1.6×
Yadea Q1 / DE3 sodium-ion, Chinese retail	136,380–144,648	1.8–1.9×
Tailg M6, Chinese retail	161,184	2.1×
Cheapest Pakistani e-bike (Evee Flipper)	97,000	1.2×
Pakistani volume band	150,000–250,000	1.9–3.2×
Yadea T5 in Pakistan	**253,500**	**3.3×**
Honda ICON e:	419,900	5.4×

TWO READINGS, AND THE SECOND IS THE UNCOMFORTABLE ONE

A Yadea sold in Pakistan retails at 3.3× what Yadea's own audited accounts say it leaves the factory for. That multiple is freight, duty, sales tax, withholding, clearing, assembly and distribution margin — the whole opportunity and the whole tariff exposure in a single number.

And **China's own retail — PKR 124,000–161,000 — lands inside Pakistan's volume band and well below the T5's PKR 253,500**, while already carrying Chinese distribution margin. **Pakistani buyers pay a large premium over Chinese buyers for the same class of machine.**

That premium is the importable margin. **It is also exactly what a domestic assembler holding the 1% CKD rate is positioned to compete away** — so it is the competitive risk as much as the opportunity.

SECTION 9

Price catalogues

Pakistan — the ladder

TIER	PKR	REPRESENTATIVE MODELS
Below the CD 70	97,000–149,900	Evee Flipper 97,000 · REVOO A04 100,000 · REVOO Y04 119,000 · Metro X8 129,000 · Evee Mito 137,000
Volume band	150,000–250,000	~35 models from 10 brands — Jolta, Pakzon, Okla, Evee, REVOO, MS Jaguar, Yadea M3/Ruibin
Mid	250,000–400,000	Yadea T5 253,500 · EcoDost ED-70 255,000 · Yadea T5L 290,000 · ELFA EV-125 345,000 · Vlektra Retro 379,000
Premium	400,000+	**Honda ICON e: 419,900** · Yadea Velax 444,000 · Okla OMAX 599,000 · Okla OKG 819,000 · Yadea Keeness 1,400,000
Petrol benchmark	**159,900**	**Honda CD 70** — 55–60 km/l

Full list: 13 brands, ~90 models. Roughly 35 models cluster within PKR 100,000 of each other — **nothing is differentiated on price.**

India — mainstream, ex-showroom

MODEL	TOP SPEED	RANGE	BATTERY	INR
Lectrix SX25	25 kmph	60 km	1.34 kWh	54,999
Ampere Reo	25 kmph	80 km	1.44 kWh	71,999
OLA S1 X	125 kmph	320 km	5.2 kWh	89,999
Honda QC1	50 kmph	80 km	1.5 kWh	90,707
TVS Orbiter	68 kmph	158 km	3.1 kWh	1,02,459
Bajaj Chetak	80 kmph	153 km	3.5 kWh	1,19,038

MODEL	TOP SPEED	RANGE	BATTERY	INR
TVS iQube	82 kmph	212 km	5.3 kWh	1,19,250
Honda Activa e	80 kmph	102 km	3 kWh	1,20,130
Ather Rizta	80 kmph	160 km	3.5 kWh	1,20,657
Ather 450X	90 kmph	161 km	3.5 kWh	1,53,902

THE STRUCTURAL GAP BETWEEN THE TWO MARKETS

Indian mainstream scooters carry **2.5–5.3 kWh and do 102–212 km**. Pakistan's carry **1.58–1.8 kWh and do 65–90 km. Honda sells a 102 km Activa e in India and a 65 km ICON e: in Pakistan** — same manufacturer, same segment, materially worse product.

India's floor is a **25 kmph low-speed class needing no licence or registration. Pakistan has no equivalent category** — creating or qualifying for one would open a tier beneath everything currently sold.

China — verified price points

BRAND / TIER	CNY	NOTE
Yadea ex-works ASP — e-bicycle AUDITED	**RMB 1,415**	Across 16.3m units
Yadea ex-works ASP — e-scooter AUDITED	**RMB 1,878**	The industry cost basis
Tailg — promotional	2,999	1200W · 72V 21Ah graphene · 60 km
Yadea Q1 / DE3	3,299–3,499	**Sodium-ion — no lithium**
Tailg M6	3,899	600W · 60V 23Ah graphene · 80 km+
Yadea G6 / T9	12,285–12,994	Mid tier
Yadea Fierider / Y1S	27,950–31,785	Premium

China's mainstream retails at CNY 2,999–3,899, close to the audited ex-works ASP — domestic distribution margin is thin. The gap between that and Pakistani retail is freight, duty, tax and margin. ⚠ **The cheap Chinese tier runs on graphene, a lead-acid derivative** — the price advantage and the 8–13-month fleet failure mode arrive together.

SECTION 10

Decision gates

Criteria written *before* the research, so they could not bend to it. Thresholds K1/K2 set by the principal; K5/K8 reasoned from structure.

GATE	STATUS	POSITION
K1 · Capital	CLEARED	PKR 50m ceiling — not binding for a fleet entry needing ~200 machines
K2 · Attention	CLEARED	30 hrs/week — not binding; **income clause remains untested**
K3 · Supply	OPEN	Yadea's territory taken; Huaihai's MOQ is 20× the ceiling. Answerable only by asking
K4 · Import ability	OPEN	Whether an LC can be opened at required volumes — a bank conversation
K5 · Battery backstop	AT RISK	No enforceable warranty chain identified. India shows failure here **can cost the right to sell**
K6 · Landed cost	LARGELY CLOSED	Duty rates, tariff line, kit prices, **battery prices and a dated FX rate all now in hand — FOB PKR 110,987.** Only the battery's own HS line and DG freight remain
K7 · Policy	FIRING	**Concessions sunset 30 June 2027**; successor unapproved; subsidy tapers. India proves the downside
K8 · Parts & service	DECISIVE	India's largest player destroyed by this from 50% share. **For a fleet, uptime is the product**
K9 · Volume	FAVOURABLE	Fleet needs under 1% of 50,000+ known riders

GATE	STATUS	POSITION
K11 · Defensible position	HARD	~88% incumbent share, 500+ dealers. **Only answerable via a segment the incumbent will not serve**

SECTION 11

What this dossier does not establish

- **Five supplier RFQs were sent on 27 August 2026** — three CKD kit makers and both battery suppliers — but **no reply has been received yet**, so every supplier figure here remains a published listing, not a quote. **No bank, delivery platform or regulator has been contacted at all.**
- **What a charging or swap station actually costs is unknown.** The import duty is 0% and unconditional, but 8504.4030 is a *static converter* line and a swap cabinet with racking and battery management may classify elsewhere. Nothing here covers the electricity connection — where the real cost and delay usually sit.
- **The correct 12-digit sub-code for a 72V traction pack.** The rates captured sit on the "portable application" sub-code; a traction pack may fall elsewhere. Heading-level rates should hold, but the sub-code must be confirmed before filing. **Lead-acid's own statutory rate was not captured**, so the cost of the excluded chemistry is unsized.
- **Dangerous-goods freight on UN38.3 lithium is uncosted**, and the IOCO quota process for the 0% battery routes — timeline, cost, feasibility — is entirely unknown. IOCO is a different body from EDB.
- **Three referenced instruments have not been read:** SRO 693(I)/2006 (the ACD on localised parts), SRO 929(I)/2024 sr.3(xi) (whose own record in the government portal is self-contradictory on its expiry), and SRO 655(I)/2006.
- **Whether a delivery platform will deduct rental at source.** The fleet model's collection mechanism depends on it entirely.
- **How EDB applies an ICE-era facility schedule to an electric vehicle.** It gates the only favourable duty rate and no published standard exists to self-assess against.
- **Whether the 200-unit CBU pilot survived AIDEP's expiry.** The most useful provision found, and possibly extinct.
- **Pakistani market volumes are press-reported.** PAMA's own data files were not obtained — and PAMA counts members only, which excludes almost every electric maker. A penetration figure computed on that basis was withdrawn from this research.
- **The FX rate is indicative, not a dealing rate.** Forex.pk's own disclaimer says so. An importer pays a bank's spread on top, and no forward rate has been obtained — an import business carries FX exposure across a 25–35 day sea leg.
- **The TCO has not been re-run at a lower petrol price.** Now that Pakistan reprices daily, a sustained crude fall transmits to the pump within days. The saving is large enough to survive it, but the sensitivity is unrun.
- **Swap-station capex and network density have not been costed.**
- **No Pakistani brand publishes unit sales.** The catalogues show what is offered, not what sells.
- **China cannot be enumerated** — hundreds of makers, thousands of SKUs, brand sites geo-blocked, pricing behind marketplace sessions, heavy white-labelling.

> ONE METHODOLOGICAL NOTE WORTH KEEPING
>
> Two conclusions in this research were **reversed by primary sources after being drawn from trade press** — a market-share ranking that could not be reconciled with audited filings, and a penetration figure computed across mismatched populations. **Where a filing and the press disagreed, the filing won.** Any figure here still carrying a PRESS grade should be treated as provisional.

SECTION 12

Sources

Audited company filings

- Yadea Group Holdings — Announcement of Annual Results, year ended 31 December 2025 (1585.HK)
- Aima Technology Group — Annual Report FY2025, 315pp (603529.SH)
- Jiangsu Xinri E-Vehicle — Annual Report FY2025, 190pp (603787.SH)

- Luyuan Group Holding — Annual Report FY2024 incl. five-year summary (2451.HK)
- Niu Technologies — Form 20-F FY2025, SEC EDGAR CIK 1744781

Pakistani primary regulation

- New Energy Vehicles Policy 2025-30 — Ministry of Industries & Production
- FBR — Salient Features, Budget 2026-27 (Customs Act 1969)
- SRO 656(I)/2006, updated to 30 June 2021 — auto OEM facility criteria
- Auto Industry Development and Export Policy (AIDEP) 2021-26
- NEPRA tariff determination — SRO 279(I)/2026
- **THE FIFTH SCHEDULE TO THE CUSTOMS ACT, 1969 (IV OF 1969)** — FBR, 87pp, FY2025-26 edition, read in full (Part-V(A) Table-II and Part-I serials 143 and (d))
- **Finance Act 2026** — Gazette of Pakistan Extraordinary, Part I, 26 June 2026, 256pp; s.3(14) and the Second Schedule (the substituted Fifth Schedule), read from the scanned pages
- **The Provincial Motor Vehicles Ordinance, 1965** (W.P. Ord. XIX of 1965) — full text
- **Sindh Excise, Taxation & Narcotics Control Department** — motor vehicle registration fee schedule
- **Balochistan Excise** — Schedule of Taxes and Fees, Jan 2026 (the kW→cc conversion rule)
- **S.R.O. 1063(I)/2026**, 30 June 2026 — additional customs duty, superseding SRO 1151(I)/2025
- **Trade Information Portal of Pakistan** — Pakistan Single Window: statutory duty, cess and measure records for 8507.6000, 8711.6040 and 8711.6060
- Board of Investment — Fifth Schedule EV concession detail, PCT 8711.6040

Industry, market and pricing

- China Chamber of Commerce for the Motorcycle Industry — 2025 rankings (⚠️ irreconcilable with audited filings; see Section 6)
- PAMA — membership roster (basis of the withdrawn penetration figure)
- PakWheels — per-brand model and price listings, Pakistan
- BikeWale — electric scooter and electric bike listings, India
- Alibaba — CKD kit listings, PKR-localised, seller-declared
- Global Sources — Dongguan Rishengzhi New Energy, 72V 30Ah LiFePO4 pack listing
- Made-in-China — Hunan CTS Technology and Koyosonic Power, 72V pack listings with quantity tiers
- Maaz Electronics (Pakistan) — 72V 30Ah LiFePO4 retail listing
- Forex.pk — interbank rate table, dated 27 August 2026 15:56 PST (indicative, not a dealing rate)
- OGRA / Petroleum Division — daily petrol notifications, August 2026, via ARY News, Daily Pakistan, Mashriq TV
- Indian regulatory and market reporting — FAME-II, PM E-DRIVE, SFIO proceedings

> RESTRICTED MATERIAL — NOT REPRODUCED HERE
>
> Official bilateral trade statistics (UN Comtrade) were used in the underlying analysis and informed the conclusions on import volumes and unit values. **The licence governing that data prohibits redistribution**, so the figures are held privately in the project workspace and are deliberately excluded from this dossier.

Compiled 27 August 2026. This is a research dossier, not an offer or a recommendation to invest. Prospective readers should conduct independent verification of every load-bearing figure.